# Hai-Xia Ma

**Nagoya University, Japan** | **ma.haixia.k8@s.mail.nagoya-u.ac.jp** | **Homepage Link**

## Research Interests

Active Galactic Nuclei; Infrared Galaxies; Computational methods; High-Redshift Galaxies; Quasar-Galaxy Pairs; Machine Learning; Galactic Dynamics; Galaxy Clusters; Galaxy Formation and Evolution; Dark Matter; Fuzzy Dark Matter; Milky Way Structure; James Webb Space Telescope; ALMA; Subaru; Gaia; LAMOST; SDSS.

## Education & Appointments

**National Astronomical Observatory of Japan & Nagoya University**, Oct. 2026 - Sep. 2027; Nagoya, Japan
ALMA Project Postdoctoral Researcher

- Supervisor: Prof. Hideki Umehata

**Nagoya University**, Doctor of Science Oct. 2022 - Sep. 2026; Nagoya, Japan

- Advisor: Prof. Tsutomu T. Takeuchi
- Thesis: Data Driven Inference of Multi-Scale Gravitational Mass Distributions

**Nagoya University**, Master of Science Oct. 2020 - Sep. 2022; Nagoya, Japan

- Advisor: Prof. Tsutomu T. Takeuchi
- Thesis: *Understanding the Origin of Multi-Scale Mass Distribution in the Universe: from Stars to Galaxies*

**Yunnan Observatories, Chinese Academy of Sciences**, July. 2019 - Sep. 2020; Yunnan, China
Research Assistant

- Advisor: Prof. Xiao-Bo Dong

**University of Science and Technology of China**, Sept. 2015 - Jun. 2019; Hefei, China
Bachelor of Astronomy

- Advisor: Prof. Xiao-Bo Dong
- Thesis: *Tests of Gravity Theories in the Andromeda Galaxy.*

## Research Visit

**Steward Observatory, University of Arizona**, Visiting Researcher Sept. 2025; Tucson, USA

- Host: Prof. George H. Rieke, Prof. Jianwei Lyu & Dr. Yongda Zhu
- Collaborative research on JWST mid-IR AGN diagnostics and spatially resolved SED fitting.

**Steward Observatory, University of Arizona**, Visiting Researcher July. 2025; Tucson, USA

- Host: Prof. George H. Rieke, Prof. Jianwei Lyu & Dr. Yongda Zhu
- Collaborative research on ANN emulators for JWST/MIRI photometry–informed SED fitting.

## Publications

**ORCiD:** 0000-0002-5237-9433

**ADS link:** ADS publication list

## First Author and Co-first Author

**[1] sOPTICS: A Modified Density-Based Algorithm for Identifying Galaxy Groups/Clusters and Brightest Cluster Galaxies** Feb. 2025

**H.-X. Ma**; T. T. Takeuchi; S. Cooray; Y. Zhu

*Monthly Notices of the Royal Astronomical Society*, 537(2), 1504–1517.

DOI: 10.1093/mnras/staf115

**[2] How Close Dark Matter Haloes and MOND Are to Each Other: Three-Dimensional Tests Based on Gaia DR2** Mar. 2023

Y. Zhu; **H.-X. Ma (co-first author)**; X.-B. Dong; Y. Huang; T. Mistele; B. Peng; Q. Long; T. Wang; L. Chang; X. Jin

*Monthly Notices of the Royal Astronomical Society*, 519, 4479–4498.

DOI: 10.1093/mnras/stac3483

## Co-author

**[3] Event-driven formulation of cosmic structure formation via Morse-Cerf theory** Oct. 2026

T. T. Takeuchi; **H.-X. Ma**; Y. Ogane

*Physical Review D*.

DOI: 10.1103/k2zn-96wd

**[4] A Potential Link between Nuclear Winds and Cold Gas Outflows on Kiloparsec Scales in Reionization-era Quasars** Mar. 2026

Y. Zhu; M. J. Rieke; L. C. Ho; Y. Sun; G. H. Rieke; F. Yuan; T. J. L. C. Bakx; G. D. Becker; J. Yang; E. Bañados; M. Bischetti; C. Cain; X. Fan; Y. Fudamoto; S. Hashemi; R. Ikeda; Z. Ji; X. Jin; W. Liu; Y. Liu; J. Lyu; **H.-X. Ma**; T. T. Takeuchi; H. Umehata; F. Wang; W. L. Tee

*The Astrophysical Journal*, 1000(2), 312.

DOI: 10.3847/1538-4357/ae4ed1

**[5] Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$** Dec. 2025

Y. Zhu; E. Egami; X. Fan; F. Sun; G. D. Becker; C. Cain; H. Chen; A.-C. Eilers; Y. Fudamoto; J. M. Helton; X. Jin; M. Pudoka; A. J. Bunker; Z. Cai; J. B. Champagne; Z. Ji; X. Lin; W. Liu; **H.-X. Ma**; Z. Ma; R. Maiolino; G. H. Rieke; M. J. Rieke; P. Rinaldi; Y. Sun; W. L. Tee; F. Wang; J. Yang; M. Yue; J. Zhang

*The Astrophysical Journal*, 995(1), L5.

DOI: 10.3847/2041-8213/ae1f8e

**[6] A Promise for the JWST Era: Massive Black Holes Directly Collapsed from Wave Dark Matter Haloes, and Star Formation in and around Their Accretion Flows** Aug. 2025

X. Dong; Y. Zhu; M. Rieke; G. Rieke; X. Li; P. Behroozi; **H.-X. Ma**; R. Meng; Z. Mao; Z. Sun

*arXiv preprint*, arXiv:2508.09258 [astro-ph].

DOI: 10.48550/arXiv.2508.09258

**[7] Discovery of a Unique Close Quasar–DSFG Pair Linked by a [C II] Bridge at $z = 5.63$** Nov. 2024

Y. Zhu; T. J. L. C. Bakx; R. Ikeda; H. Umehata; G. D. Becker; C. Cain; J. B. Champagne; X. Fan; Y. Fudamoto; X. Jin; **H.-X. Ma**; Y. Sun; T. T. Takeuchi; W. L. Tee

*Research Notes of the AAS*, 8(11), 284.

DOI: 10.3847/2515-5172/ad91ad

**[8] Probing Ultralate Reionization: Direct Measurements of the Mean Free Path over $5 < z < 6$** Oct. 2023

Y. Zhu; G. D. Becker; H. M. Christenson; A. D'Aloisio; S. E. I. Bosman; T. Bakx; V. D'Odorico; M. Bischetti; C. Cain; F. B. Davies; R. L. Davies; A.-C. Eilers; X. Fan; P. Gaikwad; M. G. Haehnelt; L. C. Keating; G. Kulkarni; S. Lai; **H.-X. Ma**; A. Mesinger; Y. Qin; S. Satyavolu; T. T. Takeuchi; H. Umehata; J. Yang

*The Astrophysical Journal*, 955, 115.

DOI: 10.3847/1538-4357/aceef4

**[9] Characterizing and Understanding Galaxies with Two Parameters** Oct. 2023

S. Cooray; T. T. Takeuchi; D. Kashino; S. A. Yoshida; **H.-X. Ma**; K. T. Kono

*Monthly Notices of the Royal Astronomical Society*, 524, 4976–4995.
DOI: 10.1093/mnras/stad2129

## Observation Experience & Proposal Involvement

### Selected projects

ALMA - Cycle 11 (PI: Zhu) – Galaxy over/under-densities around IGM transmission at z=5.7: a robust constraint on reionization

ALMA - Cycle 9 (PI: Zhu) – The Mean Free Path of Ionizing Photons at z = 5.6: A Robust Constraint on Reionization

## Technical Proficiency

**Programming:** C++, C, Python, Julia, CUDA, PyTorch, SQL, etc.

**Software:** CASA, CARTA, Gadget-4, Gadget-3, GALAXY, FreeFem++, etc.

**Language:** English (professional), Japanese (conversational), Chinese (native).

## Awards, Fellowships, & Grants

THERS Make New Standards Program for the Next Generation Researchers Fellowship, Nagoya University — FY2024

Student PI of Innovation Training Programs for Undergraduate, Chinese Academy of Sciences — FY2018

Outstanding Student Scholarship, University of Science and Technology of China — FY2018

Outstanding Student Scholarship, University of Science and Technology of China — FY2015

## Presentations

**Pixel-by-Pixel Galaxy SED Fitting Powered by Neural Network Emulators** — Dec. 2025
Oral Presentation: First Star First Galaxy Workshop 2025. Fukui, Japan

**Accelerating SED Analysis: An ANN Emulator for Dust-Obscured AGNs** — Nov. 2025
Oral Presentation: ALMA Workshop 2025. Kanazawa, Japan

**Evidence for an Oblate Core-like Dark Matter Halo in the Milky Way: Constraints from Gaia DR3 and LAMOST DR8** — Aug. 2025
Poster Presentation: 11th Galaxy Evolution Workshop. Nagoya, Japan

**Revisiting The Small-Scale Crises Of ΛCDM: New Perspectives And Insights** — Feb. 2025
*__Invited Talk__: 152nd Tsukuba Uchu Forum. Tsukuba, Japan

**Addressing the Crises of the ΛCDM Model on Galactic Scales.** — Nov. 2024
Oral Presentation: First Star First Galaxy Workshop 2024. Nagano, Japan

**A Slightly Oblate Dark Matter Halo with a Core-like Inner Density Profile Revealed by Fine-Stratified Stellar Populations from Gaia DR3.** — Oct. 2024
Oral Presentation: Kashiwa-no-ha Dark Matter and Cosmology Symposium. Tokyo, Japan

**Couplings between dark matter and baryonic matter on galactic scales in Gaia DR3.** — Oct. 2024
Oral Presentation: COSMO'24. Kyoto, Japan

**Unveiling the Shape of the Milky Way's Dark Matter Halo: Resolving the Small-Scale Challenges of CDM.** — Oct. 2024
Talk at NAOJ Subaru Telescope Group. Tokyo, Japan

**Density-based clustering algorithm for galaxy group/cluster identification.** — Sep. 2024
Poster: First Results from the SRG/eROSITA All-Sky Survey: From Stars to Cosmology. Garching, Germany

**Investigating the Temporal Evolution of Large-Scale Structure in the Universe via Topological Approaches.** Jul. 2023
Poster: TDA Week 2023. Kyoto, Japan

**Couplings between dark matter and baryonic matter on galactic scales: the radial vs. vertical.** Jun. 2023
Oral Presentation: Celebrating 40 years of Milgromian dynamics and charting the road ahead. St Andrews, UK

**Galaxy Cluster and Group Finding via Unsupervised Clustering.** Feb. 2023
Oral Presentation: 9th Galaxy Evolution Workshop. Kyoto, Japan

**Cluster Finding via Unsupervised Machine Learning.** Feb. 2022
Poster: 8th Galaxy Evolution Workshop. Online

**Tests of MOND and CDM on Galactic-scale.** Feb. 2021
Poster: 7th Galaxy Evolution Workshop. Online